Project Endgame: 40 kW 4WD Electric Race Platform

Bence Visztra | Mechatronics & Power Electronics Portfolio | Sheet 1 of 6

System Overview & Requirements

- **Objective:** Ground-up engineering of a 40 kW peak 4WD electric racing vehicle designed around high-grip 10×4.5-5 racing kart slicks.
- **Torsion-Box Billet Chassis:** Monocoque-style rails CNC-milled from 6082-T6 and 7075-T6 aluminum, eliminating torsional flex under $> 1.5\,G$ lateral cornering loads.
- **Environmental & Thermal Packaging:** Sealed ($\sim$IP65) structural enclosure with laser-cut acrylic/aluminum covers, cable glands, and integrated heatsinks for direct conductive cooling of 4x high-current motor controllers (2G300 / R200S). Houses a 20S6P high-current battery pack capable of $\sim$500–600 A peak current.
- **Track Validation:** 1st Place finish at the 2026 Budapest Championship debut; holder of the 11.96 s T-Race world record.

Specifications	**Powertrain:**	4WD Inrunner BLDC, 40 kW Peak, 84 V (20S6P)
	Chassis Mass:	$\sim$10 kg structural aluminum frame
	Protection:	Sealed bellypan, glands, loomed phase harnesses
	Manufacturing:	In-house 3-axis CNC CAM, sheet metal, laser cutting

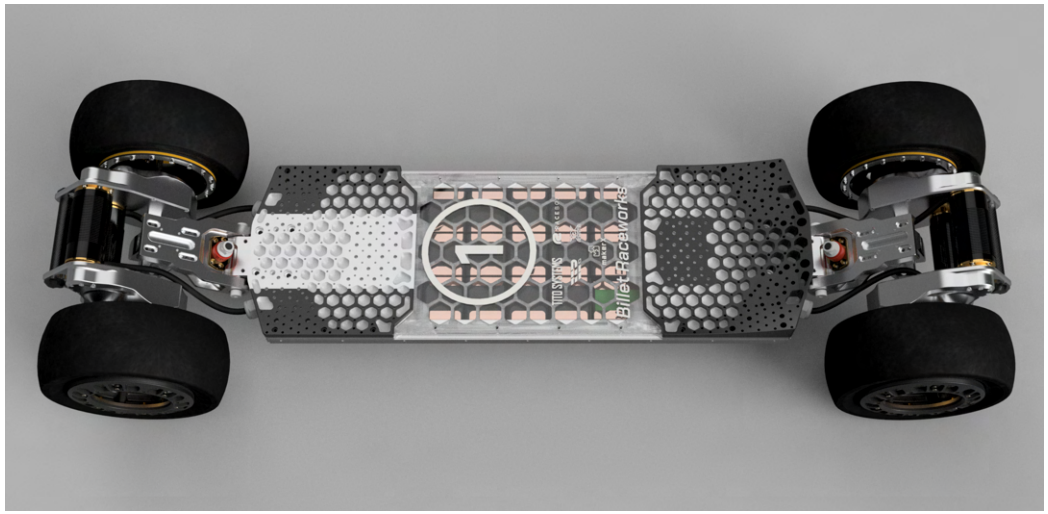


Fig 1.1: Complete CAD assembly showing chassis design, battery, and swingarms.



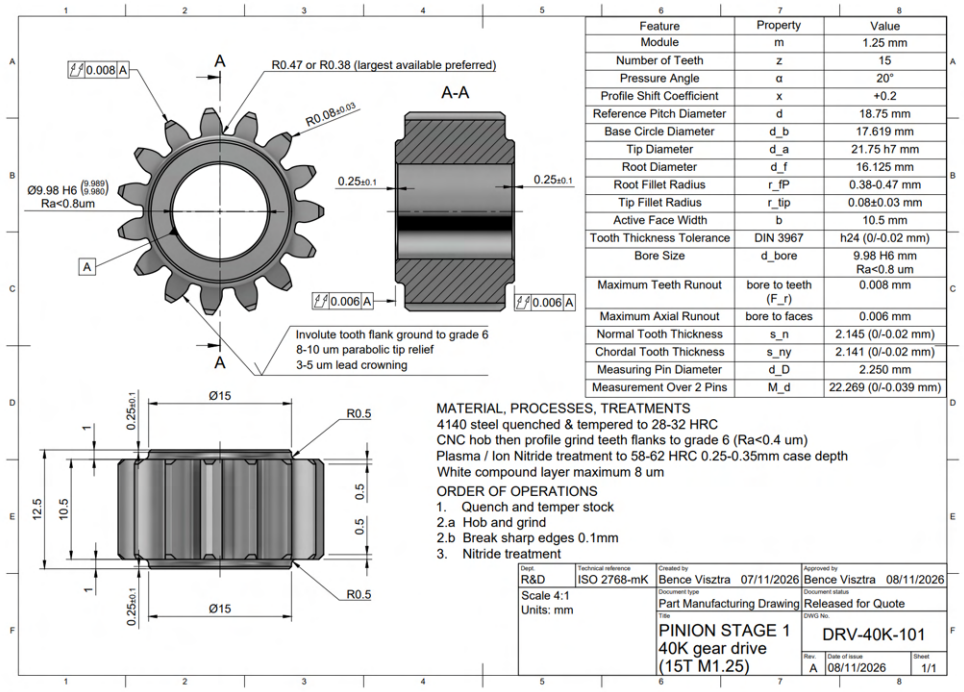
Fig 1.2: Assembled physical prototype configured with 10-inch kart slicks.

Two-Stage Reduction Drive (40k RPM)

- **Kinematics & Sizing:** Engineered a compact two-stage spur gear reduction to couple 40,000 RPM, 17 kW inrunner motors (7 Nm peak torque) to 10-inch wheels within strict 240–256 mm axle hanger width constraints.
- **Fatigue & Material Selection:** Sized gear modulus (M1.25 / M1.75) and tooth width for infinite fatigue life using 4140 quenched, tempered, and plasma-nitrided steel (58–62 HRC case hardness).
- **Production Drawing Package:** Generated complete supplier-ready RFQ drawings specifying involute profile shift ($x = +0.2$), DIN 3967 tooth tolerances, parabolic tip relief (8–$10\ \mu$m), lead crowning, and H6 bearing fits.
- **3-Gear SLM Idler Variant:** Designed an alternative compact 3-gear idler drive utilizing Selective Laser Melting (SLM) for the gears and in-house CNC milling for the housing.



Fig 2.2: 3-gear SLM downhill idler prototype (81 mm wheel).



Feature	Property	Value
Module	m	1.25 mm
Number of Teeth	z	15
Pressure Angle	α	20°
Profile Shift Coefficient	x	+0.2
Reference Pitch Diameter	d	18.75 mm
Base Circle Diameter	d_b	17.619 mm
Tip Diameter	d_a	21.75 h7 mm
Root Diameter	d_f	16.125 mm
Root Fillet Radius	r_fP	0.38-0.47 mm
Tip Fillet Radius	r_tip	0.08±0.03 mm
Active Face Width	b	10.5 mm
Tooth Thickness Tolerance	DIN 3967	h24 (0/-0.02 mm)
Bore Size	d_bore	9.98 H6 mm Ra<0.8 um
Maximum Teeth Runout	bore to teeth (F_r)	0.008 mm
Maximum Axial Runout	bore to faces	0.006 mm
Normal Tooth Thickness	s_n	2.145 (0/-0.02 mm)
Chordal Tooth Thickness	s_ny	2.141 (0/-0.02 mm)
Measuring Pin Diameter	d_D	2.250 mm
Measurement Over 2 Pins	M_d	22.269 (0/-0.039 mm)

Fig 2.1: Production drawing `DRV-40K-101` (Stage 1 Pinion) released for external gear grinding & nitriding quote.



Fig 2.3: Internal two-stage kinematics.



Fig 2.4: Sealed billet aluminum housing.

Swingarm Kinematics & Billet Fabrication

- **Elastomer Swingarm Suspension:** Designed a rigid pivot swingarm assembly utilizing preloaded angular contact bearings, 12.9-grade M12 kingpins, and custom aluminum-bronze wear inserts. Emulates longitudinal deck flex (20% reduction in turning radius under 5 mm elastomer compression mid-corner) while maintaining zero torsional slop.
- **Multi-Setup Indexed Milling:** Programmed 4-setup CAM toolpaths in Fusion 360 to machine 875 mm structural rails on a 300 mm travel CNC bed using pass-through enclosure doors.
- **Closed-Loop DFM Iteration:** Identified a datum stack-up error during Op 2 shoulder-bolt counterboring on the first article (referencing outer stock contour vs. Op 1 bore center). Revised WCS probing routine to reference internal bore geometry directly, achieving ± 0.02 mm concentricity.
- **Composite/Sheet-Metal Wheel Hubs:** Engineered a cost-optimized rim widening system combining 12 laser-cut aluminum retaining rings with an FDM core loaded purely in compression, reducing unit cost by 60% vs. SLS printing.

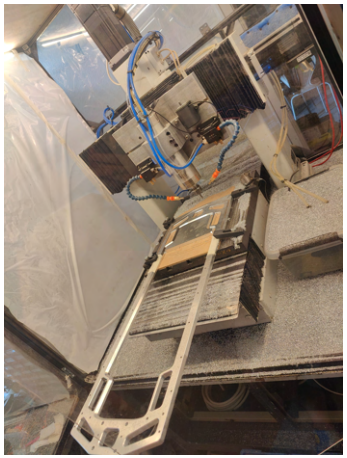


Fig 3.1: 4-setup indexed CNC milling of 875 mm rails.

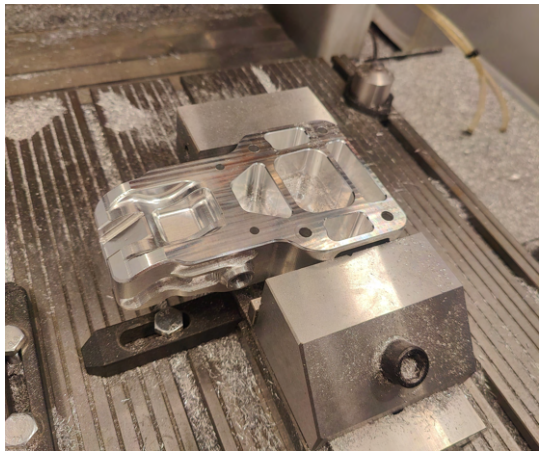


Fig 3.2: Multi-axis workholding of swingarm base in vice.

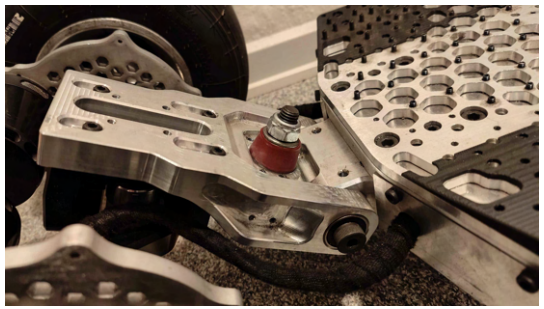


Fig 3.3: Machined billet swingarm assembly installed on chassis.

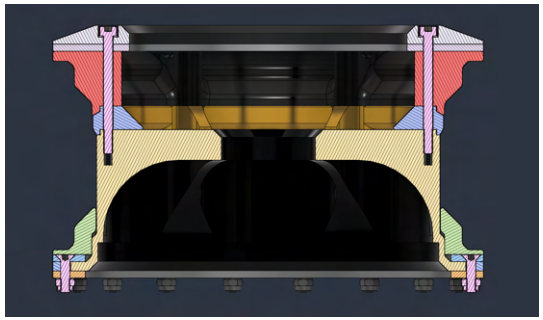


Fig 3.4: Hybrid laser-cut alu / compression-polymer rim section.

High-Current Battery Packs (Up to 84V / 40 kW)

- **Tabless Cell Architecture (20S6P & 18S6P):** Engineered and built custom high-discharge packs utilizing BAK 45D and Reliance RS50 tabless 21700 cells (70–100 A pulse capability per cell).
- **Conductor & Busbar Sizing:** Replaced standard nickel strips with spot-welded 0.3 mm pure copper interconnects and multi-run 10/8/6 AWG high-strand silicone leads to minimize I^2R thermal losses at > 250 A continuous battery current.
- **Mechanical Isolation & Insulation:** Designed custom interlocking ASA 3D-printed cell holders for thermal resilience and vibration isolation; integrated fishpaper insulation, Kapton layering, and strain-relieved balance harnesses.
- **Smart BMS & Telemetry:** Integrated 22S JBD Smart BMS units with multi-point NTC thermal probes and real-time cell monitoring.

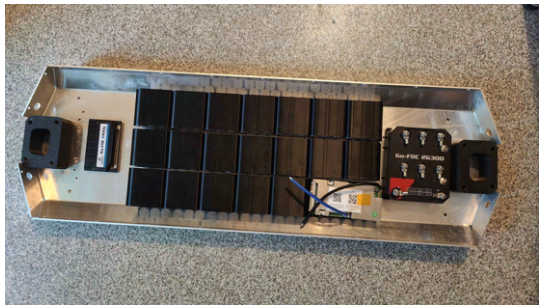


Fig 4.1: Custom ASA interlocking cell holders (20S6P) in enclosure.

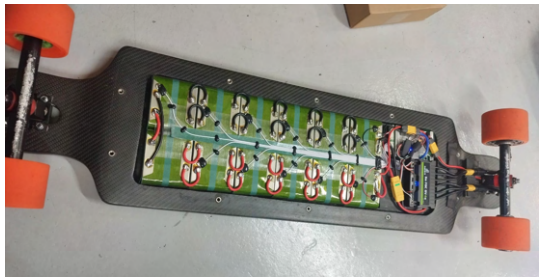


Fig 4.2: 12S4P flexible pack with precision balance routing.

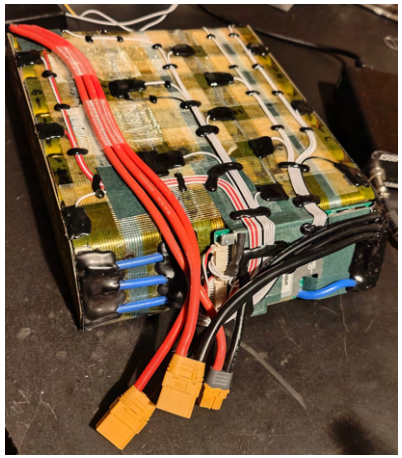


Fig 4.3: 18S6P BAK 45D pack with welded copper busbars.

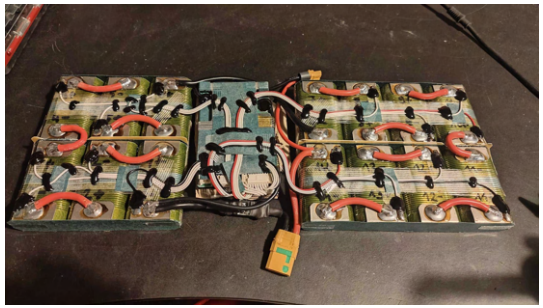


Fig 4.4: 14S2P Molicel P45B high-discharge battery pack.

800 kg CNC Mill Retrofit & Structural Redesign

- **Closed-Loop Servo Actuation:** Upgraded axis drives to 660 W 72 V NEMA 34 closed-loop JMC servos. Integrated an electromagnetic power-off holding brake on the Z-axis ballscrew to prevent spindle drop on power loss. Manually tuned the servo motor's PID loop to ∼0.02 mm following error at maximum acceleration.
- **EMC, Grounding & Bring-Up:** Diagnosed and resolved intermittent step-loss anomalies by isolating control MCU power supplies, resolving floating ground loops, and tuning angular contact ballscrew bearing preloads (AXNBT 1560).
- **Deflection Measurement & Structural CAD Upgrade:** Measured frame compliance under cutting loads with dial indicators; engineered a full X/Z axis structural upgrade transitioning from round linear guides to preloaded linear profile rails to support an Automatic Tool Changer (ATC) spindle and longer Z travel.



Fig 5.1: 800 kg mill during dial-indicator tramming.



Fig 5.2: Upgraded 660 W 72 V NEMA 34 servos and enclosure.

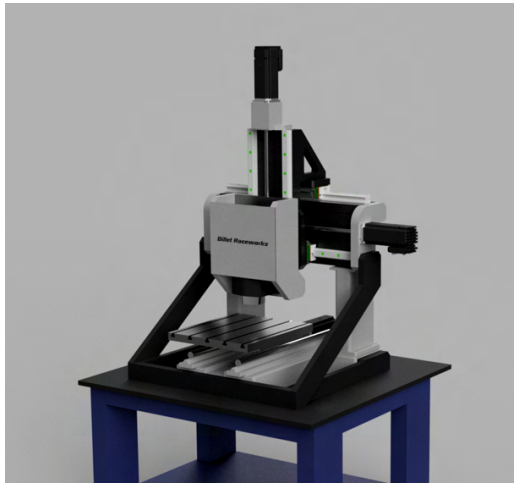


Fig 5.3: Full CAD assembly of profile-rail X/Z axis structural redesign.

Custom Inverters, Diagnostics & Vehicle Modelling

- **STM32 FOC Motor Controller (ESC):** Designed a 3-phase PMSM inverter PCB featuring low-inductance DC-link busbars, isolated gate drivers, and inline shunt sensing. Implemented discrete Field-Oriented Control (SVPWM, Park/Clarke transforms, PI current/velocity loops) on STM32. Identified SPI-DMA bottlenecks on external ADCs during bench V&V and scoped architectural revision using internal ADCs.
- **EIS Analyzer:** Built an Electrochemical Impedance Spectroscopy (EIS) board for sinusoidal cell excitation and equivalent circuit modelling.
- **CasADi / MATLAB Lap Time Optimizer:** Developed a numerical vehicle dynamics solver importing real-world RaceChrono GPS/IMU telemetry to compare against mathematically optimal trajectories—identifying acceleration and jerk losses down to 0.05 s resolution.

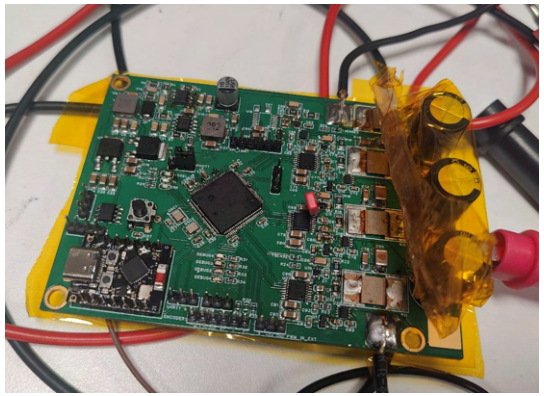


Fig 6.1: Custom STM32 3-phase FOC ESC PCB.

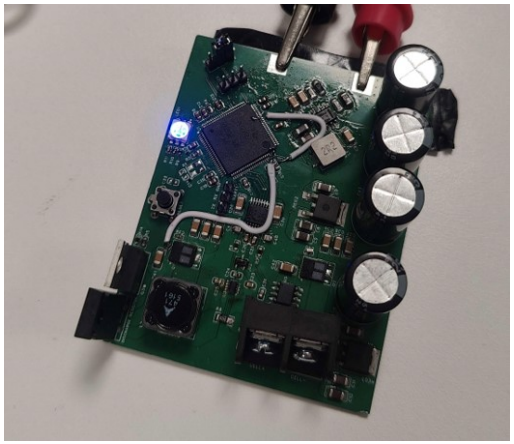


Fig 6.2: Custom EIS cell impedance analyzer PCB.

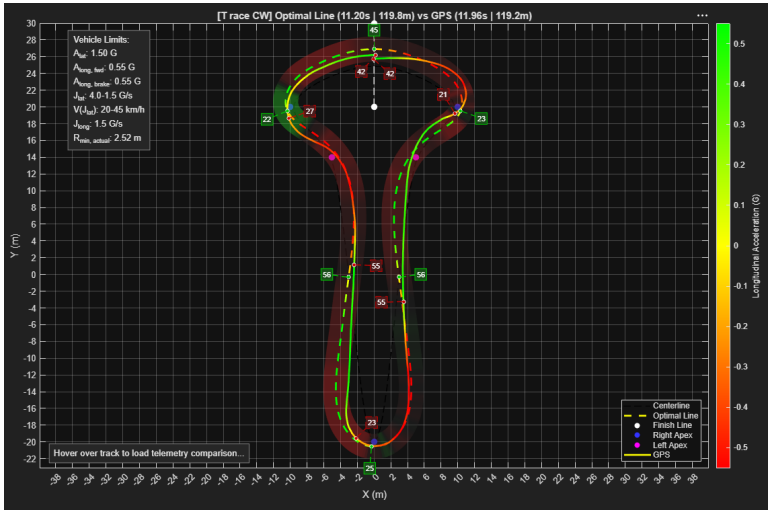


Fig 6.3: MATLAB / CasADi trajectory optimization plot comparing 11.96 s world-record GPS log against theoretical vehicle limits.